

COURSE CODE: DCIT 208

**D3:Product Backlog, UX Prototype and
Sprint 1 Plan**

PROJECT TITLE: ScentHub

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GitHub Classroom Team Repository URL:

<https://github.com/selasi07/ScentHub->

Relevant Commit Hash:

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1.Product Backlog

1.1 Introduction

The Product Backlog is a well structured list of all the work required to develop the ScentHub system. It was created from the Software Requirements Specification (SRS) prepared in Deliverable D2. The backlog breaks down the system into smaller user stories that can be planned, assigned, implemented, and tested. Each story includes a priority level, story point estimate, owner, dependencies, risks, and acceptance criteria to support effective sprint planning and project management.

1.2 Epics

The major features of the ScentHub system were identified from the Software Requirements Specification (SRS) and organized into epics. An epic is a high-level feature that represents a major area of functionality within the system. Each epic is further divided into smaller user stories that can be developed and completed within one or more sprints.

The project epics identified for ScentHub are:

- Authentication
- Product Management
- Inventory Management
- Sales Management
- Customer Management
- Reporting
- Dashboard

1.3 Refined Product Backlog

The refined product backlog was developed by breaking each epic into smaller user stories. Each story was estimated using the Fibonacci story point scale (1, 2, 3, 5, and 8), assigned a priority based on business value, and supported with ownership, dependencies, risk notes, and acceptance criteria. This refined backlog serves as the foundation for sprint planning and project implementation.

Table 1 presents the refined product backlog for the ScentHub system. The backlog was developed by decomposing the project epics into smaller user stories and assigning priorities, story point estimates, owners, dependencies, risks, and acceptance criteria

Table 1

Story ID	Epic	User Story	Story Points	Priority	Owner	Dependencies	Risk	Acceptance Criteria
ST1	Authentication	As a user,I want to log into the system so i can securely access the system.	3	Must	Backend Developer	None	Unauthorized access to the system	User enters valid credentials and is redirected to the dashboard. Invalid credentials display an error message.
ST2	Authentication	As a user, I want to log out the system so that my account is secure after use	1	Must	Backend Developer	ST1	User forgets to log out	User is successfully logged out and returned to the login page.
ST3	Product Management	As a business owner, I want to add a product so that it can be sold through the system.	3	Must	Frontend Lead	ST1	Incorrect product information.	Product details are validated, saved and displayed in the product list.
ST4	Product Management	As a user, I want to edit product details so that product information is accurate	3	Must	Frontend Lead	ST3	Invalid product updates	Updated product details are correctly saved and displayed.
ST5	Product Management	As a business owner, I want to delete a product so that unavailable products are removed from the system.	2	Must	Frontend Lead	ST3	Accidental deletion	Confirmation is requested before deletion and the product is removed successfully.

ST6	Product Management	As a user, I want to view all products so that I can see the available inventory.	1	Must	Frontend Lead	ST3	Products fail to load	All available products are displayed correctly.
ST7	Product Management	As a user, I want to search for a product so that I can quickly find it.	2	Should	Frontend Lead	ST6	Incorrect search results	Relevant products are displayed based on the search keyword.
ST8	Inventory Management	As a business owner, I want to view current stock levels so that I know the quantity of each product available.	1	Must	Database Developer	ST3	Incorrect inventory records	Stock quantities are displayed accurately for all products.
ST9	Inventory Management	As a business owner, I want to update stock quantities so that inventory remains accurate.	3	Must	Database Developer	ST8	Incorrect stock updates	Updated stock quantity is saved and displayed correctly.
ST10	Inventory Management	As a business owner, I want to record newly received stock so that inventory stays up to date.	3	Should	Database Developer	ST9	Duplicate stock entries	New stock is successfully added and inventory is updated.

ST11	Inventory Management	As a business owner, I want to receive low stock alerts so that I can restock products before they run out	2	Could	Database Developer	ST8	No alert sent.	Low-stock products are clearly identified when the quantity is reduced.
ST12	Sales Management	As a sales attendant, I want to search for a product so that I can quickly begin a sale.	2	Must	Backend Developer	ST7	Incorrect search results	Matching products are displayed based on the search entered.
ST13	Sales Management	As a sales attendant, I want to confirm the customer so that the sale is recorded for the correct customer.	2	Must	Backend Developer	ST12	Wrong customer selected	The selected customer details are displayed and confirmed before checkout
ST14	Sales Management	As a sales attendant, I want to record a sale so that every transaction is stored in the system.	5	Must	Backend Developer	ST13	Transaction failure.	The completed sale is saved successfully with all transaction details.
ST15	Sales Management	As a sales attendant, I want to confirm a sale so that the transaction is finalized correctly.	2	Must	Backend Developer	ST14	Duplicate confirmation	A confirmation message is displayed and the sale is finalized.

ST16	Sales Management	As the system, I want to automatically update stock after a sale so that inventory remains accurate.	3	Muat	Database Developer	ST14	Stock not updated	Stock quantity decreases automatically after a successful sale
ST17	Customer Management	As a business owner, I want to add a customer so that customer details can be stored for future transactions.	3	Should	Backend Developer	ST1	Incorrect customer information	Customer details are validated and saved successfully.
ST18	Customer Management	As a business owner, I want to edit customer information so that customer records remain accurate.	2	Should	Backend Developer	ST17	Invalid customer updates	Updated customer information is saved and displayed correctly.
ST19	Customer Management	As a business owner, I want to view all customers so that I can access customer records when needed.	1	Should	Frontend Lead	ST17	Customer records fail to load	All registered customers are displayed successfully.
ST20	Customer Management	As a business owner, I want to search customers so that I can quickly locate their details	2	Could	Frontend Lead	ST19	Incorrect search results	Matching customer records are displayed correctly.

ST21	Reporting	As a business owner, I want to view sales reports so that I can monitor business performance.	3	Must	Business Analyst / QA	ST14	Incorrect report data	Sales report displays accurate transaction records.
ST22	Reporting	As a business owner, I want to view inventory reports so that I can monitor stock levels.	3	Should	Business Analyst / QA	ST9	Incorrect inventory data	Inventory report displays current stock accurately.
ST23	Reporting	As a business owner, I want to export reports so that I can share or print business records.	5	Could	Business Analyst / QA	ST21, ST22	Export failure	Reports are successfully exported in the selected format.
ST24	Dashboard	As a business owner, I want to view the dashboard so that I can quickly see key business information.	2	Must	Frontend Lead	ST1	Dashboard fails to load	Dashboard loads successfully after login.
ST25	Dashboard	As a business owner, I want to view a sales summary so that I can monitor daily sales performance.	2	Should	Frontend Lead	ST21	Incorrect sales summary	Dashboard displays an accurate sales summary.

ST26	Dashboard	As a business owner, I want to view a low-stock summary so that I know which products need restocking.	2	Should	Frontend Lead	ST22	Incorrect stock summary	Dashboard displays all products below the stock threshold.
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1.4 Story Point Estimation

Story points were estimated using the Fibonacci sequence (1, 2, 3, 5, and 8) to represent the relative effort and complexity of each user story. Smaller stories were assigned lower values, while more complex stories requiring additional development or testing received higher estimates. No user story exceeded 8 story points, in accordance with the project requirements.

1.5 Backlog Prioritization

The backlog was prioritized using the MoSCoW prioritization technique (Must, Should, Could, and Won't). Core features required for a functional ScentHub system, such as authentication, product management, inventory management, and sales recording, were classified as Must. Features that improve usability but are not essential for the initial release were classified as Should or Could. No features were classified as Won't for this deliverable because the team intends to review them after client feedback and Sprint 1 planning.

2.Backlog Traceability

2.1 Introduction

Backlog traceability ensures that every user story in the product backlog can be traced back to the corresponding requirements and use cases identified in the Software Requirements Specification (SRS). This helps confirm that all client requirements are addressed, identifies stories outside the project scope, highlights areas requiring client clarification, and draws attention to features involving security or sensitive data.

2.2 Traceability Matrix

Table 2 shows the relationship between the product backlog, functional requirements, and use cases defined in the SRS.

Story ID	User Story	SRS Requirement	Use Case
ST1	Login	FR1	UC1 Login
ST2	Logout	FR1	UC1 Logout
ST3	Add Product	FR2	UC2 Manage Products
ST4	Edit Product	FR2	UC2 Manage Products
ST5	Delete Product	FR2	UC2 Manage Products
ST6	View Product	FR2	UC2 Manage Products
ST7	Search Product	FR2	UC2 Manage Products
ST8	View Stock	FR3	UC3 Manage Inventory
ST9	Update Stock	FR3	UC3 Manage Inventory
ST10	Receive Stock	FR3	UC3 Manage Inventory
ST11	Low Stock Alerts	FR3	UC3 Manage Inventory
ST12	Search Product for sale	FR4	UC4 Record Sale
ST13	Confirm Customer	FR4	UC4 Record Sale
ST14	Record Sale	FR4	UC4 Record Sale
ST15	Confirm Sale	FR4	UC4 Record Sale
ST16	Automatic Stock Update	FR4	UC4 Record Sale
ST17	Add Customer	FR5	UC5 Manage Customers
ST18	Edit Customer	FR5	UC5 Manage Customers
ST19	View Customers	FR5	UC5 Manage Customers
ST20	Search Customer	FR5	UC5 Manage Customers
ST21	View Sales Report	FR6	UC6 Generate Reports
ST22	View Inventory Report	FR6	UC6 Generate Reports
ST23	Export Reports	FR6	UC6 Generate Reports
ST24	View Dashboard	FR7	UC7 Dashboard
ST25	View Sales Summary	FR7	UC7 Dashboard
ST26	View Low Stock	FR7	UC7 Dashboard

2.3 Deferred Stories

Although all identified user stories are included in the product backlog, some lower-priority features may be deferred if time constraints arise during development. These stories are not required for the first sprint and may be implemented in later iterations.

Story ID	Story	Reason for Deferral
ST11	Low Stock Alerts	Can be added after the core inventory features are complete.
ST20	Search Customers	Customer management is not essential for the initial prototype.
ST23	Export Reports	Advanced reporting feature planned for a later sprint.
ST25	Sales Summary Dashboard	Enhancement feature after core reporting is complete.
ST26	Low Stock Summary Dashboard	Depends on inventory reporting and dashboard functionality.

2.4 Stories Requiring Client Clarification

Some user stories require additional clarification from the client before implementation to ensure the delivered system meets business expectations.

Story ID	Clarification Needed
ST17	What customer details are mandatory?
ST21	Which sales report format does the client prefer?
ST22	What inventory statistics should be displayed?
ST23	Which export formats should be supported(PDF, Excel)
ST24	Which information should appear on the dashboard home screen?

2.5 Security and Sensitive Stories

Several backlog items involve authentication, business records, and customer information. These stories require additional attention to security, including proper authentication, authorization, input validation, secure password handling, and protection of customer and sales data.

Story ID	Security Concern
ST1	Secure authentication and password protection
ST2	Secure session management
ST14	Prevent duplicate or fraudulent sales
ST16	Prevent unauthorized inventory updates
ST17	Protect customer personal information
ST21	Restrict access to business reports

3. UX Sketches and Prototype

3.1 Introduction

The UX prototype provides an early visual representation of the ScentHub system before implementation begins. The purpose of the prototype is to validate the application's layout, workflow, and usability with stakeholders before development. The wireframes presented in this section focus on the system's core functionality and user experience rather than visual design.

3.2 Wireframes

Figures 1–3 present low-fidelity wireframes for the major screens of the ScentHub system. The layouts were designed to represent the intended workflow and functionality before implementation.

Figure 1: Login Wireframe.

Figure 2: Dashboard Wireframe

Figure 3: Product Management Wireframe

ScentHub

Welcome Back

Email/Username

Password

LOGIN

[Forgot Password?](#)

[Create Account](#)

ScentHub Dashboard

[Home](#) | [Products](#) | [Profile](#) | [Logout](#)

Total Products
50

Orders
25

Customers
100

Add Product

Manage Products

Recent Activities
-Added new product
-Updated product details
-Deleted product

Product Management

Search Product_____

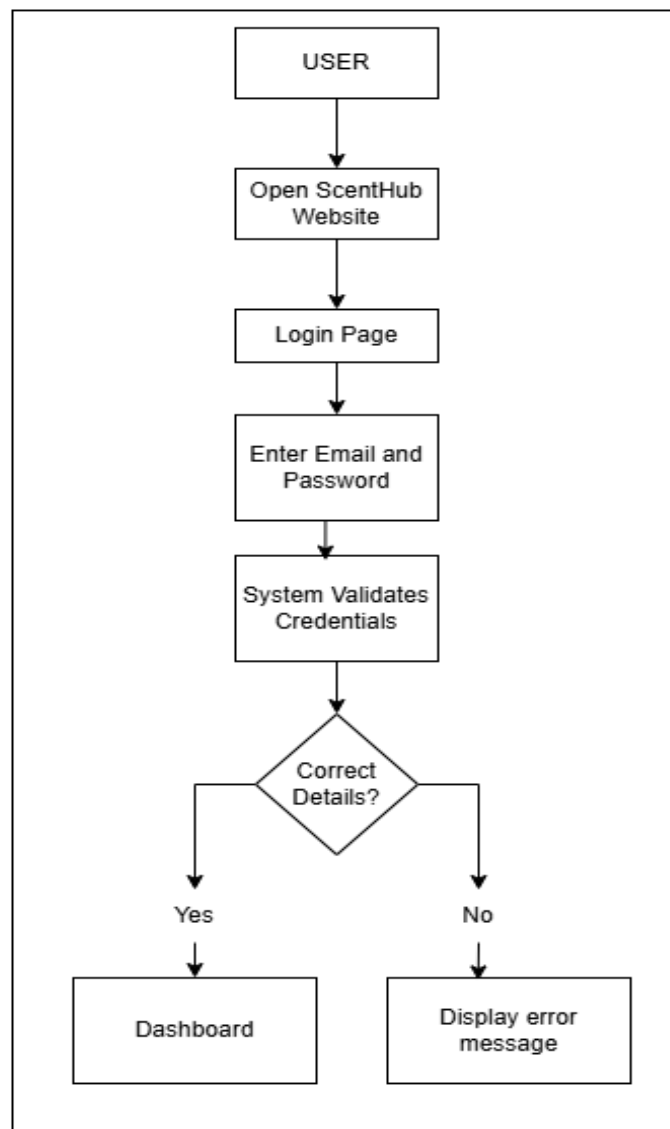
+ Add Product

Product Name	Price	Stock	Action
Perfume A	50	20	Edit Delete
Perfume B	70	15	Edit Delete

3.3 User Flows

Flow	Description
User flow 1	Login → Dashboard
User flow 2	Dashboard → Products → Add Product → Save Product
User flow 3	Dashboard → Sales → Search Product → Confirm Customer → Record Sale → Sale Confirmation

The diagram below represents possible scenarios for User Flow 1



Possible Failure: Server unavailable or incorrect login details.

Security Concern: Passwords must be encrypted.

3.4 Usability Assumptions

The prototype was designed with simplicity and ease of use in mind. Navigation is kept consistent across all screens, buttons are clearly labelled, and the number of steps required to complete common tasks is minimized. The interface assumes that users have basic computer literacy and require quick access to inventory and sales functions.

3.5 Accessibility Considerations

The interface considers accessibility by using readable font sizes, clear navigation labels, sufficient spacing between controls, and high-contrast colour combinations. Forms are designed with descriptive labels to improve usability for all users.

3.6 Client Review Evidence

The proposed wireframes were presented to the client for preliminary feedback. The client indicated that the overall workflow aligned with the intended business process and recommended keeping the interface simple and easy to navigate. Minor refinements will be incorporated during implementation based on future client feedback.

4.Sprint 1 Plan

4.1 Sprint Goal

Sprint 1 focuses on developing the core functionality required for a working prototype of the ScentHub system. The goal is to implement user authentication, basic product management, inventory viewing, and the initial sales workflow. At the end of Sprint 1, users should be able to log into the system, manage products, view stock information, and record a basic sale.

4.2 Sprint Backlog

Story ID	User Story	Story Points	Owner
ST1	Login	3	Backend Developer
ST3	Add Product	3	Frontend Lead
ST6	View Products	1	Frontend Lead
ST8	View Stock	1	Database Developer
ST9	Update Stock	3	Database Developer
ST12	Search Product	2	Backend Developer
ST14	Record Sale	5	Backend Developer
ST16	Automatically Update Stock	3	Database Developer

Total Sprint Capacity: 21 Story Points

4.3 Sprint Capacity

Based on the availability and experience of the team members, the team estimates a Sprint 1 capacity of approximately 21 story points. This workload is considered achievable within the sprint while allowing sufficient time for testing, documentation, meetings, and addressing unforeseen issues.

4.4 Story Owners

Team Member	Role
Abukari Issah Kangre	Project Manager & Frontend/UI-UX Lead
Kelly Adoboe	Database Developer, Scrum Master & Documentation Lead
Joel Asiedu	Backend Developer & Database Lead
Ishawu Abdul Manaf	UI/UX Designer, Graphics & Quality Assurance Engineer

4.5 Planned Tests

Story	Planned Test
Login	Verify login with valid credentials.
Login	Reject invalid username or password.
Add Product	Verify product is saved successfully.
View Products	Verify all saved products are displayed.
View Stock	Verify inventory quantities are correct.
Record Sale	Verify a completed sale is stored correctly.
Automatic Stock Update	Verify stock decreases after each successful sale.

4.6 Definition of Ready (DoR)

This tells when a story is ready to begin development.

A user story is considered ready when:

- ❖ The requirements are clearly understood.
- ❖ Acceptance criteria have been defined.
- ❖ Dependencies have been identified.
- ❖ The story has been estimated using story points.
- ❖ An owner has been assigned.
- ❖ The Product Manager has approved the story for development.

4.7 Definition of Done

This tells us when a story is finished.

A user story is considered Done when:

- The feature has been implemented.
- All acceptance criteria have been satisfied.
- Unit and functional tests have passed.
- Code has been reviewed by another team member.
- Documentation has been updated.
- The feature has been successfully demonstrated to the team.

5. GitHub Project Board

5.1 Introduction

GitHub Projects was used to organize, monitor, and manage the development activities of the ScentHub project. Each user story from the product backlog was created as a GitHub Issue and assigned to a project board for Sprint 1 planning. Labels, milestones, and issue assignments were used to improve collaboration and track project progress throughout the development process.

5.2 Project Board

A GitHub Project Board was created to organize all Sprint 1 tasks. The board follows a Kanban workflow consisting of the following columns:

Column	Purpose
To Do	Stories waiting to be started
In Progress	Stories currently being implemented
Done	Completed stories that satisfy the Definition of Done

5.3 Labels

Labels were created to categorize issues according to the type of work required. These labels make it easier to filter tasks and assign responsibilities.

Label	Purpose
Frontend	User interface development
Backend	Business logic implementation
Database	Database design and queries
Testing	Test development and validation
Documentation	Documentation updates
Security	Authentication and security tasks
Bug	Defects identified during development
Ai-assisted	AI-assisted planning or implementation
requirement	Requirement-related tasks

5.4 Milestone

A milestone named **Sprint 1** was created to group all user stories selected for the first sprint. This milestone allows the team to monitor progress towards the Sprint 1 goal and measure completion before the sprint deadline.

5.5 GitHub Issues

Each user story selected for Sprint 1 was created as an individual GitHub Issue. Every issue contains a clear description, acceptance criteria, assigned owner, appropriate labels, and a link to the Sprint 1 milestone.

Issue	Labels	Assigned Role	Milestone
Login	Backend, security	Backend Developer	Sprint 1
Add Product	Frontend	Frontend lead	Sprint 1
View Products	Frontend	Frontend Lead	Sprint 1
View Stock	Database	Database Developer	Sprint 1
Update Stock	Database	Database Developer	Sprint 1
Search Product	Backend	Backend Developer	Sprint 1
Record Sale	Backend	Backend Developer	Sprint 1
Automatic Stock Update	Database	Database Developer	Sprint 1

6. Conclusion

This deliverable transformed the Software Requirements Specification into a structured product backlog, established traceability between requirements and user stories, and prepared a realistic Sprint 1 plan for implementation. Low-fidelity UX prototypes were developed to visualize the system workflow and improve usability before development begins. GitHub Issues and the project board provide a clear approach for managing tasks, tracking progress, and supporting team collaboration. Overall, this deliverable provides a strong foundation for the implementation of the ScentHub system in the next phase of the project.

